Code No: 113AU **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, September - 2021
ELECTRONIC DEVICES AND CIRCUITS
(Common to EEE, ECE, CSE, IT, MCT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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Used Papers SEP 2021

1. Define and distinguish the following terms with reference to the diode:
   a) Static and dynamic resistance b) transition and diffusion capacitance
   c) Practical vs ideal diode d) Zener and avalanche breakdown mechanism. [15]

2.a) Explain the principle of operation of varactor diode.
   b) Explain the principle of working of SCR and draw its V-I characteristics and equivalent circuit diagrams. [7+8]

3.a) Draw the circuit diagram and waveforms of Bridge full wave rectifier and explain its working.
   b) A single phase full wave rectifier use semiconductor diodes. The transformer voltage is 40V rms to center tap. The load consists of a 30μF capacitor in parallel with a 100Ω resistor. The diode and transformer resistances and leakage reactance may be neglected. Assume that the power-line frequency is 50Hz. Calculate: i) dc current in the circuit ii) peak-to-peak amplitude of the ripple voltage iii) rms value of ripple voltage iv) ripple factor of the rectifier filter output. [7+8]

4.a) Draw and explain centre-tap full wave rectifier circuit and then derive its efficiency expression.
   b) Compare different types of filters. [9+6]

5.a) Explain the potential variations through the transistor under biased conditions and derive its current equation.
   b) Draw the common base configuration, input characteristics and output characteristics and identify different regions of operation. [8+7]

6.a) Show that the exact expression for h_{fb} in terms of the CE hybrid parameters is

$$h_{fb} = -\frac{h_{fe}(1-h_{re}) + h_{ie}h_{oe}}{(1+h_{fe})(1-h_{re}) + h_{oe}h_{ie}}$$

   b) Draw the equivalent circuit for the CE and CC configurations subject to the restriction that R_L=0. Show that the input impedances of the two circuits are identical. [7+8]

7.a) What is thermal stability? How to achieve in transistor circuits?
   b) Draw the voltage divider bias circuit in CE configuration and then explain the same. [7+8]

8.a) Compare the enhancement and Depletion MOSFETs in all parameters.
   b) Draw the circuit diagram of Common Source amplifier and derive the equation for voltage gain, input and output impedances. [5+10]

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